

Random Signals and Noise (NETW504)

Milestone 1 - Exploratory Data Analysis (EDA)

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Goal: Understand your approved dataset properly before any machine-learning work. This milestone is mainly EDA. Only the basic cleaning needed to make the analysis consistent is required.

Important: Python is mandatory. Use a Jupyter Notebook. 1. Dataset and target

- Dataset name and Kaggle link.
- Number of samples (rows) and columns in the original dataset.
- In one or two sentences, explain what one row represents.
- State the exact target column name and explain what it represents.
- List all target classes/labels. The approved target must have 2 to 4 classes.
- Show the count and percentage of samples in each target class.
- List the input features and mark each as numerical or categorical.
- Remove ID-like columns (for example: customer ID, row number, record ID) from the input features. Names or free-text identifiers should not be treated as useful numerical/categorical features.

2. Initial inspection and data quality

- Display the first 5 rows, dataset shape, column names and data types.
- Count exact duplicate rows.
- For every column, show the number and percentage of missing values.
- For every numerical input feature, print: count, mean, standard deviation, minimum, median, and maximum.
- Comment briefly on class balance, missing data, unusual values and any obvious data-quality issue you observe.

3. Required EDA plots - all must appear in the notebook

The following plots are mandatory. Generate them using Python and keep every plot in the submitted notebook.

Plot	Required output
Target distribution	One count/bar plot showing the number of samples in every target class.
Missing values	One bar plot of missing-value counts for all columns before filling them.
Numerical features – PDF	For every numerical input feature, plot an estimated PDF using a normalized histogram (density)
Numerical features - CDF	For every numerical input feature, plot its empirical CDF.
Correlation	Create a correlation matrix and a heatmap for the numerical input features after the basic preprocessing below.

Plot rules: Every figure must have a clear title and labeled axes. Under each group of plots in the notebook, write a short observation (1-3 sentences). Do not paste a plot without explaining what you noticed.

4. Basic preprocessing - follow these rules

This is not a full data-engineering milestone. Apply only the following fixed steps, in this order:

1. Remove exact duplicate rows.
2. If the target is missing in a row, remove that row.
3. For missing numerical input values, fill using the median of that feature.
4. One-hot encode categorical input features. Encode the target labels as integer values (for example, No=0 and Yes=1). Keep and report the mapping.

Keep both versions in your notebook: the original data used for EDA and the cleaned/encoded data produced at the end. Do not overwrite the original dataset at the beginning.

5. PDF report - maximum 6 pages

The report should be short. Do not paste every notebook output into the report. The complete code and all plots belong in the notebook. The report should summarize the important findings.

- Dataset and target: dataset purpose, shape, target name, target meaning, target classes and class percentages.
- Data quality: duplicates, missing values and the main issues found.
- Descriptive statistics: a compact table for numerical features including mean and standard deviation (and the main quartiles where useful).
- Key EDA findings: include the target distribution, correlation heatmap, and only the most informative PDF/CDF, categorical and boxplot figures. Refer to the notebook for the full set.
- Basic preprocessing summary: what was removed/filled, outlier counts and the encoding used.
- Final dataset: final number of rows and columns, and a short conclusion (approximately 5-8 lines) describing what you learned from the EDA.

Page limit: 6 pages maximum for the complete report. Keep tables and figures readable. There is no need for a separate long introduction, literature review, or appendix.

6. What to submit

- TeamName_M1_Report.pdf - maximum 6 pages.
- TeamName_M1.ipynb - complete Python notebook. It must run from the first cell to the last cell without errors and must contain ALL required plots.
- TeamName_M1_Cleaned.csv - final cleaned and encoded dataset.

Deadline: _____